

A Terahertz Metasurface for Thin Film Biosensing

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Abstract—Terahertz (THz) metasurface biosensors have attracted considerable attention in thin-film biological sensing due to their real-time and ultra-sensitive detection properties. This paper proposes a planar array of split-ring resonators (SRRs) over a dielectric substrate to study its performance for having a highly sensitive, non-label, and low-cost biosensor. By coupling the two SRRs of each unit cell, a pick in the transmission spectra occurs at 1.58 THz. Analyzing the electromagnetic field distribution and surface current flow shows the appearance of coupled electric and magnetic dipoles in the resonators. It is also shown that changing the distance between the coupled SRRs can passively tune the resonance frequency. Finally, the sensing functionality of the metamaterial is validated by simulating a human skin tissue with a refractive index of 1.6 and a thin thickness of 6 μm . The results give a remarkable theoretical sensitivity of 400 GHz/RIU.

Index Terms—THz Metamaterial, Biosensing, Coupled SRR.

I. INTRODUCTION

THz technology has some advantages, including non-invasive behavior due to low photon energy, spectral fingerprinting, penetration capabilities, and relatively good resolution, which made it a suitable option in medical spectroscopy and imaging [1].

Metamaterials are artificial structures often made of resonators with a size scale smaller than the wavelength of the incident wave. Their electromagnetic characterisations can be changed by chemical composition and each constitutive resonator's internal structure, position, and orientation. Since their first introduction [2] [3], they faced a fast development over the past years and revolutionized the ability to control electromagnetic waves. The surface version of metamaterials is called metasurfaces with the benefit of occupying less physical space with a highly compact and less-lossy structure [4].

The sensitivity of a metasurface sensor is more crucial when dealing with thin film sensing of biological samples in which the material under test does not interact with the incident light, and the changes in THz transmitted or reflected wave are negligible. The biggest challenge in thin-film sensing is to improve the sensitivity by engineering the metasurfaces. This sensitivity in the literature has been defined from the changes of resonance frequency to the refractive index change. Different types of metasurfaces have been studied from classical plasmonic-like resonances to advance type of fano-like resonance based metasurfaces [5].

The constituent of classical plasmonic metamaterials are mainly based on SRRs fabricated on a dielec-

tric/semiconductor substrate. The operation of most SRR-based sensors depends on the designs employed to control the resonances of SRRs, increasing the electromagnetic field confinement and thus increasing the interaction between the sensor and sample under test [6] [7].

In this work, we have designed a structure that consists of double SRRs to study the effect of coupling on the resonance frequency and sensing performance. The coexistence of an electric-dipole and magnetic-dipole resonance between a circular split ring resonator (CSRR) and a hexagonal split ring resonator (HSRR) leads to interference between two elements. The electric and magnetic field distribution, along with the surface current illustration, and parametric study of the coupling distance of two resonators, has been studied. The results proved the tuning function of the biosensor. In order to examine the sensing performance of thin biological samples, we have considered the specification of skin tissue with a refractive index of 1.6 and a thin thickness of 7 μm .

II. METASURFACE DESIGN

The unit cell of the presented metamaterial consists of two metallic SRRs, including a circular SRR (CSRR) and a hexagonal SRR (HSRR) on top of a dielectric layer as illustrated in Fig. 1. The metal layer is gold with a conductivity of 4.461×10^7 S/m. The material of the dielectric layer with a thickness of 5 μm is a quartz crystal with a dielectric constant of 3.75. The incident THz waves transfer from the metal to the dielectric layer. The sample is considered on the metallic patterns. The exact value of the unit cell geometries is mentioned in the caption of Fig. 1. The metamaterial resonator is simulated and optimized by the CST microwave studio based on a finite difference time domain (FDTD) solver. The electric, magnetic and open (add space) boundary conditions are considered along x , y and z direction, respectively.

III. RESULTS AND DISCUSSION

The simulated transmission of the separated and coupled CSRR and HSRR resonators have been depicted in Fig. 2. As can be seen, the resonance frequency of HSRR and CSRR resonator alone is nearly equal to 1.44 THz and 2.33 THz, respectively. Both the CSRR and HSRR are excited due to the incident wave coupling. At the frequency of 1.58 THz, a pick in the transmission spectra occurred, and radiation losses diminished. The underlying reason for this phenomenon refers

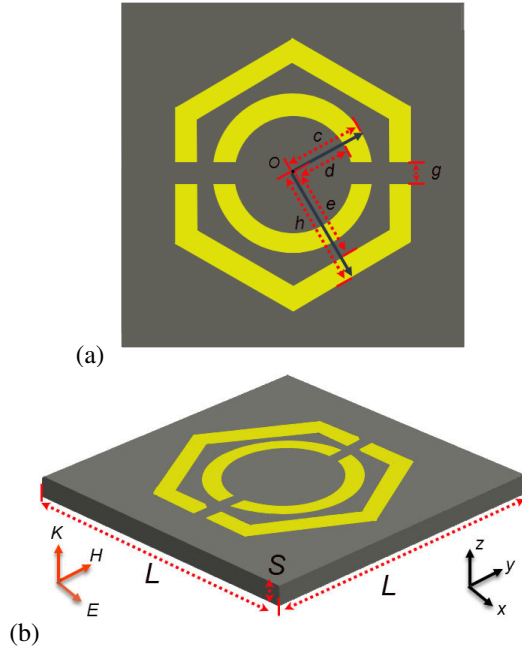


Fig. 1. (a)-(b) Top and side view of the unit cell of the proposed metamaterial with the circle center of o and below geometrical parameters : $c=20\ \mu\text{m}$, $d=15\ \mu\text{m}$, $h=35\ \mu\text{m}$, $e=30\ \mu\text{m}$, $g=5\ \mu\text{m}$, $L=90\ \mu\text{m}$ and $S=5\ \mu\text{m}$. The height of the gold layer is $200\ \text{nm}$.

to the strong coupling between two resonators. As a result, the absorption or scattering loss is significantly suppressed owing to robust coupling.

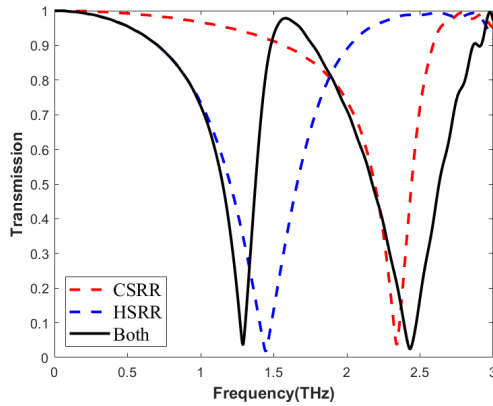


Fig. 2. Simulated transmission spectra of CSRR (red dash line) , HSRR (blue dash line) and coupling both resonators (black solid line).

In order to analyze the coupling effect, electric and magnetic field, as well as the current distribution, is illustrated at two resonance dips of f_1 and f_3 , and the transmission peak of f_2 in Fig. 3. According to Fig. 3 (a), there is a direct dipole excitation of the HSRR from the incident wave while coupling with the free space. Simultaneously, an incomplete semi-circulating current is made on each side of the SRRs and leads to a weak magnetic field, which only appears on the

side of HSRR . At the frequency of f_2 , as can be seen in Fig. 3 (b), the electromagnetic field is coupled between the HSRR and CSRR, leading to a destructive interference and suppressed loss. As can be seen, a complete circulating current is running toward the upper and bottom pair of the coupled resonators, and thus, a strong out-phase magnetic dipole is shaped between them. This field confinement makes an increased interaction of the metasurface with the material under test because any change that results in the lineshape spectra can only be associated with the change of the surrounded dielectric constant. At the frequency of f_3 , another electric dipole, related to CSRR, appears, and a weaker out-phase current distribution leads to making a magnetic field only on its close space.

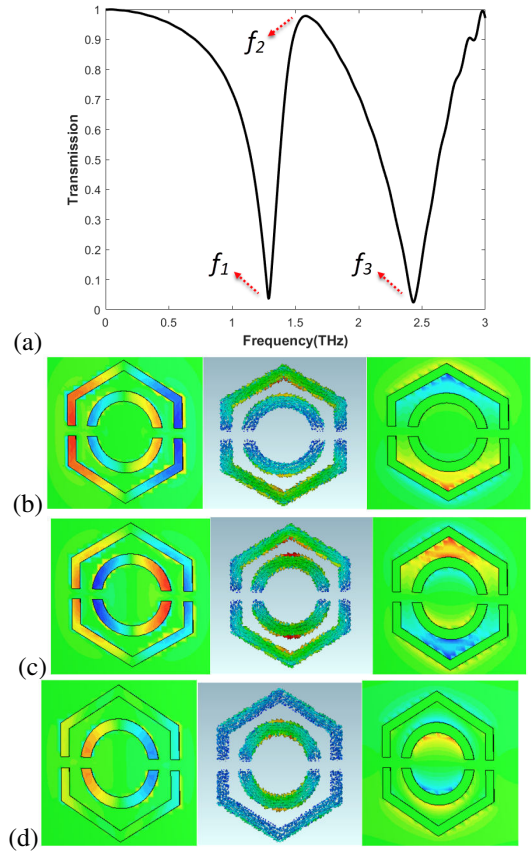


Fig. 3. Simulated transmission spectra, electric and magnetic fields distribution, and surface currents of the designed biosensor. (a) transmission (b) electric field distribution (left), surface current distribution (middle) and magnetic field distribution (right) associated with $f_1 = 1.28\ \text{THz}$. (c) same parameters with same orders associated with $f_2 = 1.58\ \text{THz}$ (d) same parameters with same orders at the resonance frequency of $f_3 = 2.43\ \text{THz}$.

The coupling between resonators is determined by their spatial separation [8]. From Fig. 1, the parameter c as the outer radius of the CSRR controls the physical distance of the gap between each side of the paired resonators. As the c increases, the distance between the two resonators decreases and strengthens the coupling. The transmission spectra of variation of this parameter are shown in Fig. 4. With the

decrease of c from $25\ \mu\text{m}$ to $20\ \mu\text{m}$, further to shifts of the lineshape to higher frequencies, the radiation and dissipation losses suppress more, and the transmission window gains a higher efficiency. This result shows the possibility of passively tuning the resonance frequency of the metamaterial from 1.27 THz to 1.58 THz.

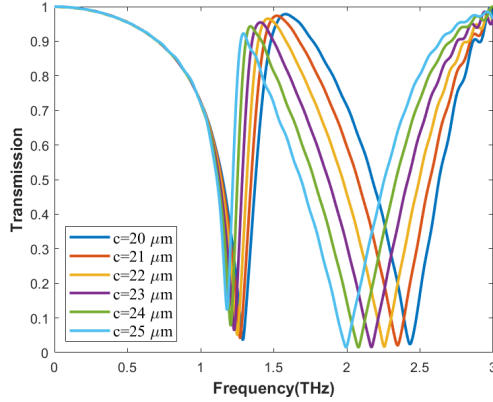


Fig. 4. Simulated transmission spectra of proposed metasurface for different values of parameter c

In order to further investigate the sensing performance of the proposed metamaterial, the theoretical sensitivity based on refractive index changes has been calculated. The theoretical sensitivity is defined by the frequency shift of the pick frequency to the change of the refractive index. To prove the bio-sensing capability of the proposed structure, we simulated an analyte with a thin thickness of $6\ \mu\text{m}$ and varied its refractive index from 1 to 1.6, which corresponds to the permittivity range of skin [9]. As can be seen in Fig. 5, a linear shift in the pick of the transmission spectra to the lower range can be observed. With changing the refractive index from 1 to 1.6, the transmission peak frequency shifts from 1.58 THz to 1.34 THz, which results in a distinctive theoretical sensitivity ($S = \Delta f / \Delta n$) of 400 GHz/RIU (RIU is the unit of refractive index).

ACKNOWLEDGMENT

This research was supported by the School of EECS, Queen Mary University of London.

IV. CONCLUSION

In this study, a metamaterial based on coupled SRRs is designed. It demonstrated that from coupling the HSRR and CSRR elements, a pick in the transmission spectra occurs at 1.58 THz in which the dissipation and absorption are suppressed and cause a high sensitivity to the variation in the external dielectric environment. This results in building a none-label, low-cost, and ultrasensitive biosensor. The electromagnetic fields plus the surface current distribution of the resonators are elaborated based on the electric and magnetic dipoles. Furthermore, the parametric study of the coupling distance of the SRRs shows the feasibility of passively tuning

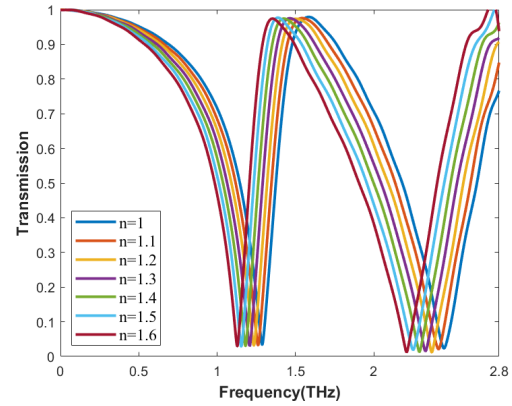


Fig. 5. Variation of transmission spectra with changing the refractive index of the analyte with the thickness of $6\ \mu\text{m}$ from 1 to 1.6.

the resonance frequency. Furthermore, its performance for thin-film biological sensing is examined by simulating a skin tissue with a refractive index of 1.6 and thin thickness of $6\ \mu\text{m}$, which gives a theoretical sensitivity of 400 GHz/RIU.

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